

PROJECT COMMISSIONING & RISK STRATEGY

The Pathway to Live Operations

Lwandile Engineering Projects (Pty) Ltd | Commissioning & Quality Protocols

Three-Phase Commissioning

A structured, methodical ramp-up prevents early infrastructure failure. We divide the launch process into three sequential validation gates before introducing any livestock capital.



Phase 1: Civil Gate

Verifying masonry curing, floor gradients, and airtight seals.

Phase 2: Energy Gate

Syncing the BESS storage with the biogas co-generation thermal manifolds.

Phase 3: Biological Gate

Sterilizing pod surfaces and dry-testing climate sensors.

Phase 1: Civil & Structural Audits

Sub-Floor & Drainage Verification

Ensure physical grading is perfect before laying down elevated plastic flooring. Soil foundations must support high payloads.

- ✓ **Drainage Slope Test:** Flooding the sub-floor channels with water to verify a true 2% gravity flow toward the digester intake.
- ✓ **Air-Tight Pod Test:** Pressurizing each of the 4 independent nursery rooms to test gasket seals and thermal barrier doors.



Structural Defense

A single leak or gradient failure under the nursery slats will pool moisture, triggering high-ammonia gas pockets and destroying piglet health.

Phase 2: Energy & Thermal Balancing

Balancing Volts and Steam

Synchronizing the solar PV arrays with the 30kW inverter, the 30kWh BESS bank, and the biogas generator exhaust.

- ✓ **BESS Grid Calibration:** Charging and discharging the battery storage over 3 continuous cycles to confirm cell balance.
- ✓ **Manifold Pressure Audit:** Testing the hot steam lines routed sub-surface beneath the nursery pod floor grids.



Thermal Integration

Verifying that engine exhaust heat exchange produces stable steam pressure, taking 100% of the heating baseline load off the battery banks at night.

Phase 3: Biological Pre-Loading

Sterilization & Dry Runs

Eradicating all pathogens and running the farm's environmental brains on an empty test cycle before loading live animals.

- ✓ **Deep-Chemical Wash:** High-pressure chemical foam washing across all walls, roofs, slats, and water drinker lines.
- ✓ **Sensor Stress Test:** Artificially dropping pod heat to verify that automated thermostats open the biogas steam lines.



Stewardship Target

Guarantees that day-old chicks and 7kg weaners enter a highly sterilized, pre-warmed, and fully monitored climate from Hour 1.

Emergency Response Protocols



BESS Fault Action

If the battery storage drops, manual switches lock the biogas generator into direct load supply, keeping nursery mats active.



Digester Failure

In case of biogas output drops, back-up LPG gas burners connect to the steam manifolds, maintaining critical room temperatures.



Water Line Breach

Emergency bypass valves disconnect local pod lines immediately, routing water through backup sanitized storage tanks.

Thermal Reserve Safety Margins

120 Hours

Water Reserve Buffer

Resilient Resource Buffers

We do not design our resource lifelines to operate near baseline limits. Our margins are engineered to survive sustained outages.

- ✓ **Hydration Defense:** Elevated water towers hold 5 full days of emergency water reserves.
- ✓ **Thermal Retention:** 50mm high-density roof insulation maintains safe indoor temperatures for up to 24 hours during a complete heating failure.

Risk Response Action Matrix

Identified Risk Event	Impact Level	Structural Mitigation Deployed	Responsible Role
Severe Highveld Storm	Moderate	90-second manual PVC winch curtain closure and dwarf wall barrier.	Section Steward
Water Dosing Failure	High	Inline UV sterilizer and backup mechanical chlorine injector.	Hydrology Lead
Nursery Pathogen Breach	Critical	Physical 4-pod air-gap isolation and localized chemical footbaths.	Biosecurity Officer

SLA & Preventative Maintenance

Protecting Infrastructure Lifespans

To secure our long-term JET investment returns, we mandate rigorous preventative maintenance protocols across all high-tech assets.

- ✓ **Solar Array Maintenance:** Weekly dust cleaning of PV panels to ensure peak winter solar recharge rates.
- ✓ **Digester Slurry Checks:** Weekly pH and temperature analysis of anaerobic digester biology to prevent chemical death.
- ✓ **BESS Ventilation:** Monthly filter cleaning of inverter housing cabinets to stop heat build-up.



INVESTMENT DEFENSE LOCKED

Ready for Groundbreak. Engineered for Life.

Lwandile Engineering Projects (Pty) Ltd

Image Sources



<https://www.aforeenergy.com/wp-content/uploads/2026/02/4-13.webp>

Source: www.aforeenergy.com